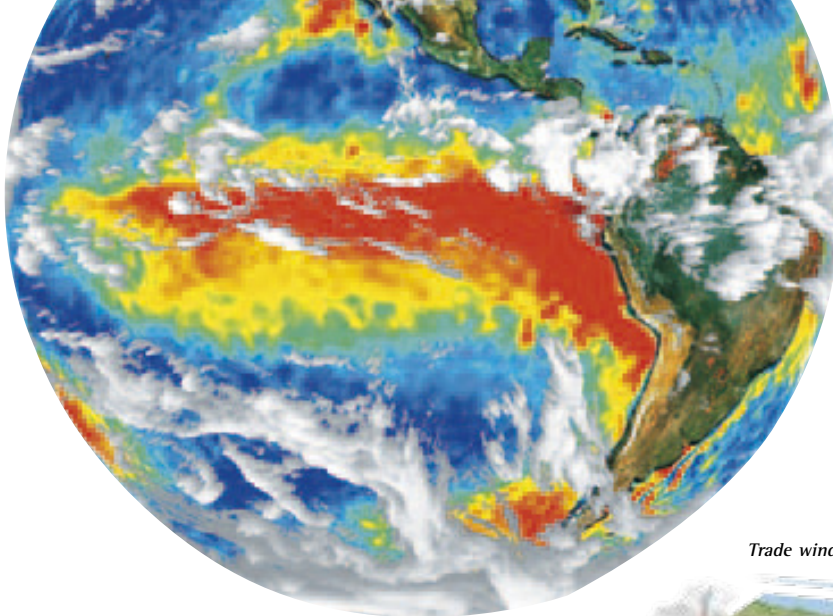


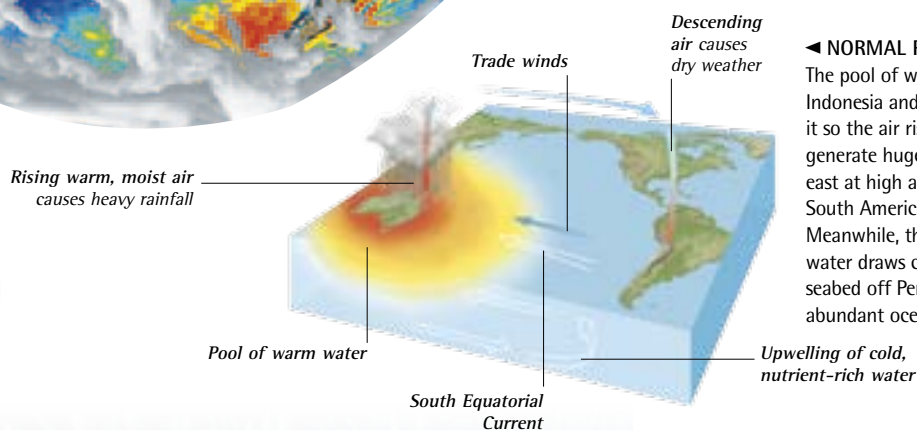


◀ EL NIÑO

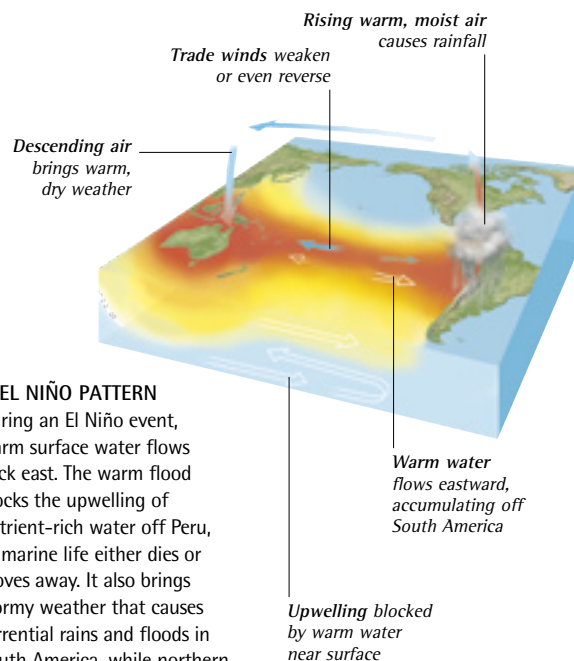
The trade winds that blow over the tropical Pacific normally drive warm surface water toward the west. But every few years, this pattern is disrupted by a change in airflow that weakens the trade winds. This allows the warm water to flood back east toward South America, as shown in this satellite-generated image. The effects of this reversal are called El Niño, which means "The Child" in Spanish, because these events always occur around Christmas.

◀ NORMAL PATTERN

The pool of warm water that builds up around Indonesia and New Guinea warms the air above it so the air rises, carrying water vapor with it to generate huge storm clouds. The rising air flows east at high altitude, then cools and descends over South America, causing cloudless, dry conditions. Meanwhile, the westward drift of warm surface water draws cool, rich water up from near the seabed off Peru, creating an upwelling zone of abundant ocean life.



seasonal shifts



▲ EL NIÑO PATTERN

During an El Niño event, warm surface water flows back east. The warm flood blocks the upwelling of nutrient-rich water off Peru, so marine life either dies or moves away. It also brings stormy weather that causes torrential rains and floods in South America, while northern Australia and the islands of the western Pacific suffer from droughts and wildfires.

◀ DEVASTATED FISHERIES

The waters off Peru are among the richest fisheries in the world, with vast shoals of anchovies and other fish that feed on the marine life of the plankton. But the plankton swarms depend on cold, nutrient-rich water reaching the sunlit surface. When El Niño events prevent this, most tiny oceanic animals die. As a result, the fish that feed on them disappear, breeding colonies of fish-eating seabirds fail, and the fisheries that sustain local economies are devastated.